SC1006
Computer Organisation and Architecture

Data Centre Storage

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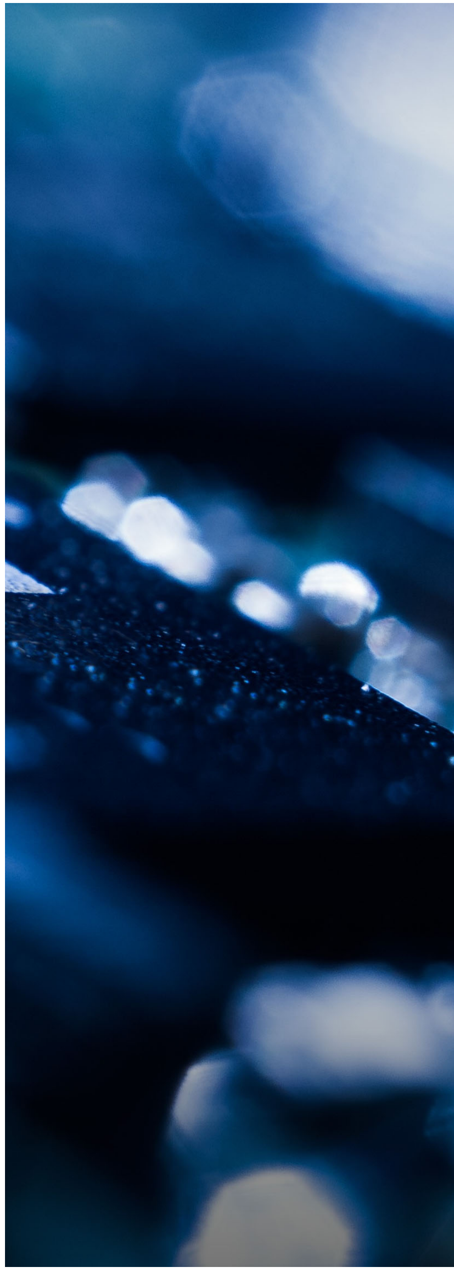


Criteria for Storage Element

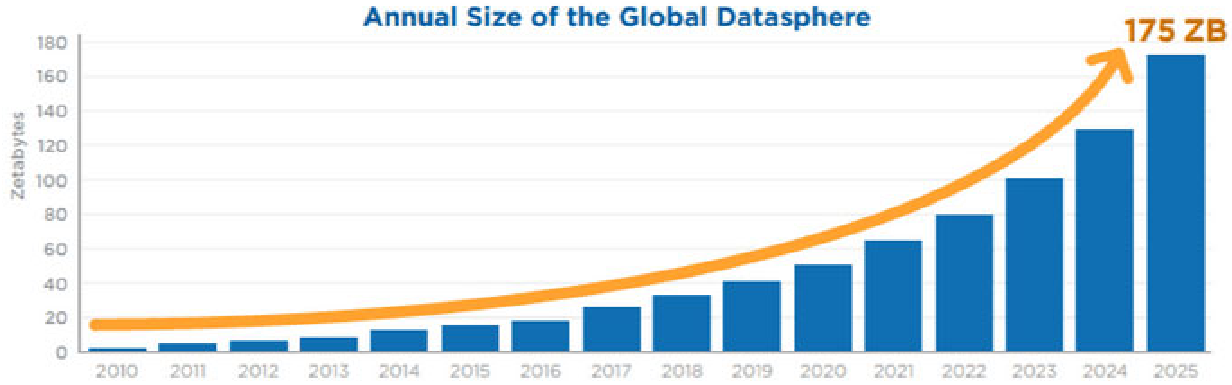
- Power consumption
  - Data centers consumed a lot of power, which not only translates to direct costs in utility and cooling measures but also limits its choice of location.
  - Centres are commonly found near natural cooling elements such as large natural water bodies which offer a low-cost and reliable source of cooling.
- Speed
  - Beneficial for caching databases and other data affecting overall application or system performance.
- Robustness
  - Tolerance to various forms of mechanical movement/interference increases reliability and reduces the need and cost of maintenance.
  - housing structure shock absorption requirement is also reduced.
- Heat production
  - The less heat generated the less cooling and power required in the data centre.
- Size
  - Data centers will be able to store more data in less space: increasing efficiency in all areas (power, cooling, etc.)

Criteria

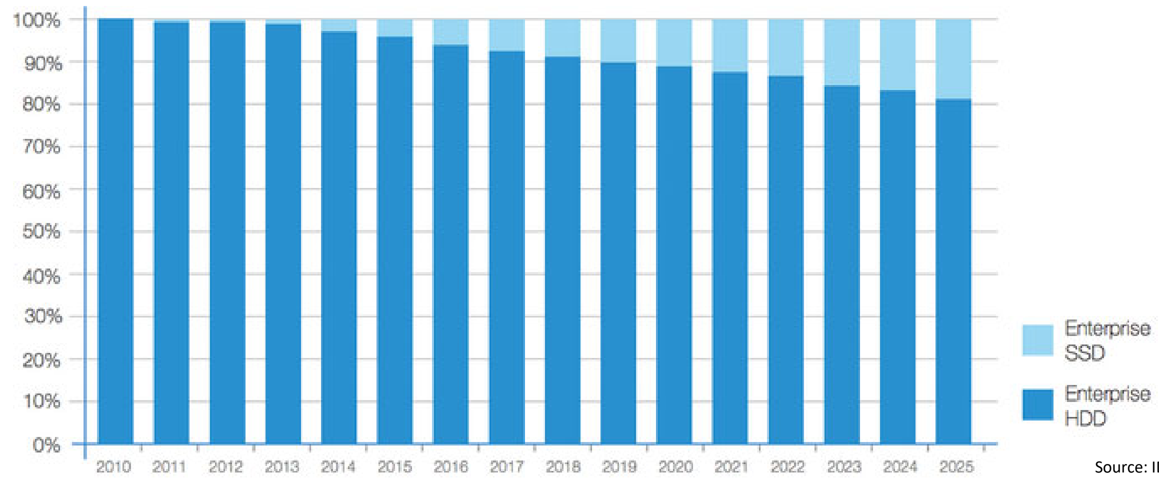
- Based on the criteria in the previous slide, SSD wins HDD hands down.
- So why is HDD still the dominant Storage in Data Centres?

COST

Global Storage Consumption and Shipment Projection

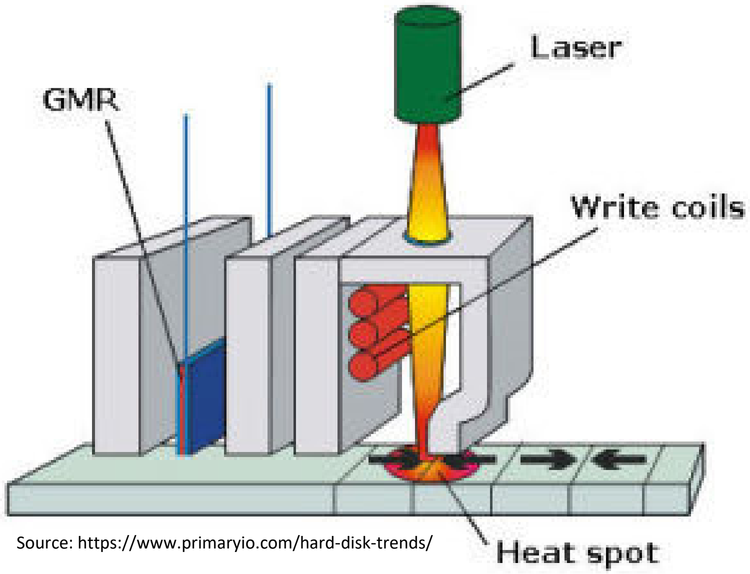


1 ZB = 10^{21} bytes

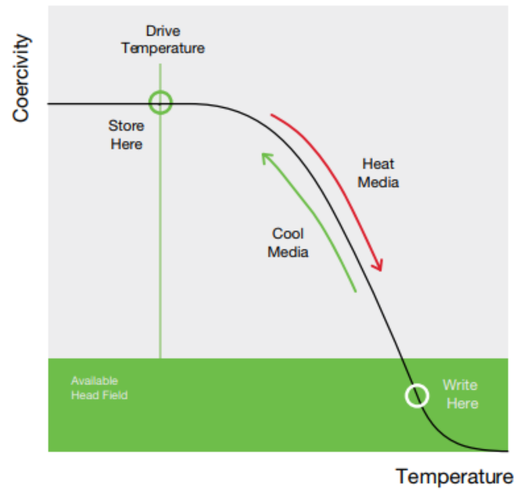


Source: IDC, Seagate.

HAMR (Heat Assisted Magnetic Recording)



Source: https://www.primaryio.com/hard-disk-trends/



Source: https://www.msp360.com/resources/blog/hamr-vs-mamr-new-hdd-technology/

- The areal density of the HDD was stagnant for a while and manufacturers had been shipping drives with 2TByte/Platter.
- But with the HAMR, the areal density is expected to increase again.
- A small laser is attached to a recording head, designed to heat a tiny spot on the disk where the data will be written. This allows a smaller bit cell to be written as either a 0 or a 1.
- Current projections are that HAMR can achieve 5 Tbpsi, enabling hard drives with capacities higher than 100 TB.

HAMR

- The major problem with packing bits so closely together on conventional magnetic media is that the data bits become unstable and may flip (0 → 1 or 1 → 0).
- To make the media maintain their stability to store bits over a long period of time, the recording media needs to have a higher coercivity.
- Higher coercivity implies the media is magnetically more stable during storage, but it would also be more difficult to change the magnetic characteristics of the media when writing.
- For that, a laser is employed to heat a tiny region of several magnetic grains for a very short time (~1 ns) to a temperature high enough to lower the media's coercive field to below that of the write head's magnetic field. This is the write process.
- Immediately after the heat pulse, the region quickly cools down and the bit's magnetic orientation is frozen in place. The data is stored in the media and would be stable due to the high coercivity of the recording media.
- See the graph in the previous slide for the visual illustration.